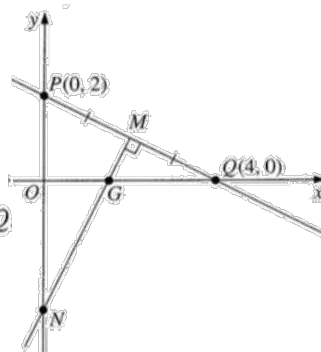


A1	2000	☒	☐
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The diagram shows the points $P(0, 2)$ and $Q(4, 0)$. The point M is the midpoint of PQ . The line MN is perpendicular to PQ and meets the x -axis at G and the y -axis at N .

- Show that the gradient of PQ is $-\frac{1}{2}$.
- Find the coordinates of M .
- Find the equation of the line MN .
- Show that N has coordinates $(0, -3)$.
- Find the distance NQ .
- Find the equation of the circle with centre N and radius NQ .
- Hence show that the circle in part (f) passes through the point P .
- The point R lies in the first quadrant, and $PNQR$ is a rhombus. Find the coordinates of R .



(a) Gradient of $PQ = \frac{0-2}{4-0} = -\frac{1}{2}$.

(b) M , midpoint of PQ is $M\left(\frac{0+4}{2}, \frac{2+0}{2}\right)$.
 $\therefore M(2, 1)$.

(c) $MN \perp PQ$,
 $\therefore$ gradient of $MN = 2$ ($m_1 m_2 = -1$).
 Equation of MN is
 $y - 1 = 2(x - 2)$
 $= 2x - 4$
 $2x - y - 3 = 0$.

(d) Substitute $x = 0$ in equation of MN .
 $2(0) - y - 3 = 0$
 $y = -3$.
 $\therefore N$ has coordinates $(0, -3)$.

(e) $NQ = \sqrt{(4-0)^2 + (0+3)^2} = 5$.
 $\therefore$ distance $NQ = 5$ units.

(f) $r = 5$. Centre is $(0, -3)$.
 $\therefore$ Equation of circle is $(x-0)^2 + (y+3)^2 = 5^2$
 or $x^2 + (y+3)^2 = 25$.

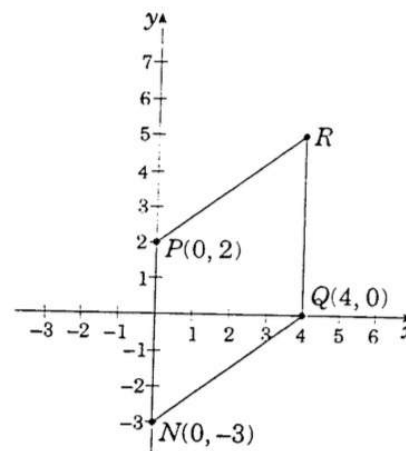
(g) $PN = 2 - (-3)$
 $= 5$
 $=$ radius of circle.
 $\therefore$ Circle passes through P .

OR Substitute coordinates of P in the equation of the circle.

LHS $= 0^2 + (2+3)^2$
 $= 25$
 $=$ RHS.

$\therefore$ Circle passes through P .

(h)



By observation: R is $(4, 5)$, since Q is 4 units to the right of PN and P is 5 units up the y -axis from N .

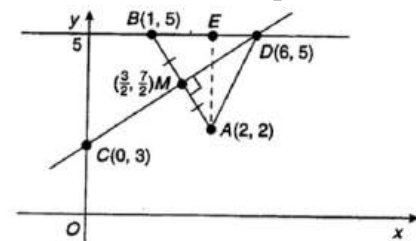


A2	2002	☒	☐
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The diagram shows two points $A(2, 2)$ and $B(1, 5)$ on the number plane.

Copy the diagram into your writing booklet.

- Find the coordinates of M , the midpoint of AB .
- Show that the equation of the perpendicular bisector of AB is $x - 3y + 9 = 0$.
- Find the coordinates of the point C that lies on the y axis and is equidistant from A and B .
- The point D lies on the intersection of the line $y = 5$ and the perpendicular bisector $x - 3y + 9 = 0$. Find the coordinates of D , and mark the position of D on your diagram in your writing booklet.
- Find the area of triangle ABD .



$$y = \frac{x}{3} - \frac{1}{2} + \frac{7}{2}$$

$$y = \frac{x}{3} + 3$$

$$\therefore 3y = x + 9$$

$$\therefore x - 3y + 9 = 0.$$

- Coordinates of M are,

$$\left(\frac{1+2}{2}, \frac{5+2}{2}\right)$$

$$\text{i.e. } \left(\frac{3}{2}, \frac{7}{2}\right).$$

$$\text{(ii) Gradient of } AB = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5-2}{1-2}$$

$$= \frac{3}{-1}$$

$$= -3$$

Since for perpendicular lines $m_1 \times m_2 = -1$,

$$\therefore \text{Gradient of perpendicular} = \frac{1}{3}.$$

$\therefore$ Equation of perpendicular bisector with gradient $= \frac{1}{3}$, passing through $M\left(\frac{3}{2}, \frac{7}{2}\right)$ is given by

$$y - \frac{7}{2} = \frac{1}{3}\left(x - \frac{3}{2}\right)$$

$$y - \frac{7}{2} = \frac{x}{3} - \frac{1}{2}$$

- As C is equidistant from A and B , it lies on the perpendicular bisector of AB , which cuts the y axis at $x = 0$:
 $\therefore 0 - 3y + 9 = 0$
 $-3y = -9$
 $y = 3$

$\therefore$ Coordinates of C are $(0, 3)$.

- At D , $y = 5$
 $\therefore x - 3(5) + 9 = 0$
 $x - 15 + 9 = 0$
 $x = 6$

$\therefore$ Coordinates of D are $(6, 5)$

$$\text{(v) Area of } \triangle ABD = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times BD \times AE$$

$$= \frac{1}{2} \times 5 \times 3$$

$$= \frac{15}{2}$$

$$= 7.5 \text{ units}^2.$$

A3	2003	☒	☐
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In the diagram, $OABC$ is a trapezium with $OA \parallel CB$. The coordinates of O , A and B are $(0, 0)$, $(-1, 1)$ and $(4, 6)$ respectively.

- Calculate the length of OA .
- Write down the gradient of the line OA .
- What is the size of $\angle AOC$?
- Find the equation of the line BC , and hence find the coordinates of C .
- Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$.
- Hence, or otherwise, calculate the area of the trapezium $OABC$.

$$\text{(i) } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(-1-0)^2 + (1-0)^2}$$

$$= \sqrt{1+1}$$

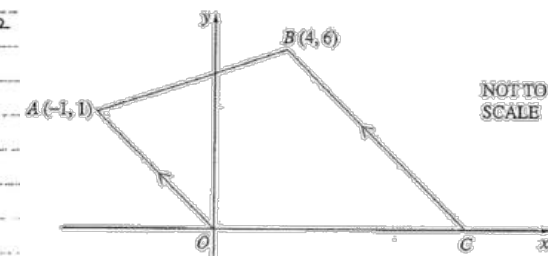
$$= \sqrt{2}$$

$$\text{(ii) } \frac{\text{rise}}{\text{run}} = \frac{-1}{1} = -1$$

$$\text{(iii) } \tan \theta = 1 = 45^\circ$$

$$45^\circ + 90^\circ = 135^\circ$$

$$\therefore \angle AOC = 135^\circ$$



$$\text{(iv) gradient of } BC = -1$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = -1(x - 4)$$

$$y = -x + 10$$

$$0 = 10 - x - y$$

$$\text{To find } C, \text{ let } y = 0. \therefore x = 10 \Rightarrow C(10, 0)$$

$$\text{(vi) } BC = \sqrt{(10-4)^2 + (-6)^2}$$

$$= \sqrt{36+36}$$

$$= \sqrt{72}$$

$$= 6\sqrt{2}$$

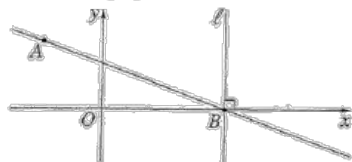
$$\text{Area} = \frac{1}{2} (\sqrt{2} + 6\sqrt{2}) \times 5\sqrt{2}$$

$$= 35 \text{ units}^2$$

A4	2004	☒	☐
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The diagram shows the points $A(-1, 3)$ and $B(2, 0)$.

The line ℓ is drawn perpendicular to the x -axis through the point B .

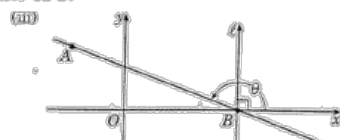


- Calculate the length of the interval AB .
- Find the gradient of the line AB .
- What is the size of the acute angle between the line AB and the line ℓ ?
- Show that the equation of the line AB is $x + y - 2 = 0$.
- Copy the diagram into your writing booklet and shade the region defined by $x + y - 2 \leq 0$.
- Write down the equation of the line ℓ .
- The point C is on the line ℓ such that AC is perpendicular to AB . Find the coordinates of C .

$$\begin{aligned} \text{(a) (i) } AB &= \sqrt{(2 - (-1))^2 + (0 - 3)^2} \\ &= \sqrt{3^2 + (-3)^2} \\ &= \sqrt{9 + 9} \\ &= \sqrt{18} \\ &= 3\sqrt{2} \text{ units.} \end{aligned}$$

$$\begin{aligned} \text{(ii) METHOD 1 } \text{Gradient } AB &= \frac{3 - 0}{-1 - 2} \\ &= \frac{3}{-3} \\ &= -1. \end{aligned}$$

$$\begin{aligned} \text{METHOD 2 } \text{Gradient } AB &= \frac{\text{rise}}{\text{run}} \\ &= \frac{-3}{3} \\ &= -1. \end{aligned}$$



Let θ be the angle between the line AB and the positive direction of the x -axis.

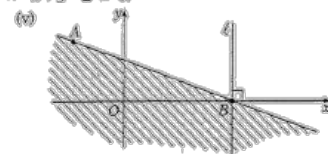
$$\begin{aligned} m &= \tan \theta \\ \text{but gradient } AB &= -1 \\ \text{so } \tan \theta &= -1 \\ \theta &= 135^\circ. \end{aligned}$$

$\therefore$ The acute angle between ℓ and line AB is $135^\circ - 90^\circ = 45^\circ$.

$$\text{(v) Equation of line } AB: y - y_1 = m(x - x_1)$$

$$\begin{aligned} \text{Given } m_{AB} &= -1, B(2, 0) \\ y - 0 &= -1(x - 2) \\ y &= -x + 2. \end{aligned}$$

$$\therefore x + y - 2 = 0.$$

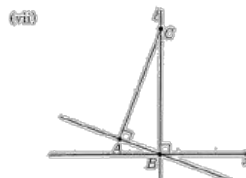


$$\begin{aligned} \text{Test } (0, 0): \text{ LHS } &= x + y - 2 \\ &= 0 + 0 - 2 \\ &= -2. \end{aligned}$$

$\therefore (0, 0)$ satisfies $x + y - 2 \leq 0$.

$\therefore (0, 0)$ lies in the region $x + y - 2 \leq 0$.

$\therefore$ Equation of line ℓ is $x = 2$.



METHOD 1
Equation of line $AC: y - y_1 = m(x - x_1)$.
Now $AC \perp AB$.
So if $m_{AB} = -1$, then $m_{AC} = 1$.
Line passes through $A(-1, 3)$.
 $y - 3 = 1(x + 1)$

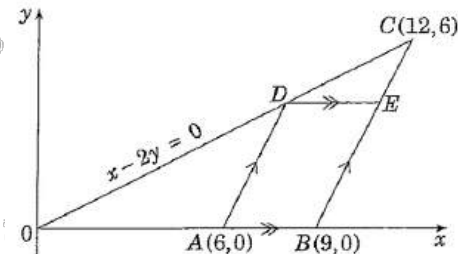
$$\begin{aligned} y &= x + 4 \\ \text{At } C, x &= 2: y = 2 + 4 \\ y &= 6. \\ \therefore \text{Coordinates are } C(2, 6). \end{aligned}$$

METHOD 2
Since C lies on the line $x = 2$, let C have coordinates $(2, y)$. Now $AC \perp AB$.
So if $m_{AB} = -1$, then $m_{AC} = 1$.

$$\begin{aligned} m_{AC} &= \frac{y - 3}{2 - (-1)} \\ 1 &= \frac{y - 3}{3} \\ 3 &= y - 3 \\ y &= 6. \\ \therefore \text{Coordinates are } C(2, 6). \end{aligned}$$

A5	2005	☒	☐
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In the diagram, A , B and C are the points $(6, 0)$, $(9, 0)$ and $(12, 6)$ respectively. The equation of the line OC is $x - 2y = 0$. The point D on OC is chosen so that AD is parallel to BC . The point E on BC is chosen so that DE is parallel to the x -axis.



- Show that the equation of the line AD is $y = 2x - 12$.
- Find the coordinates of the point D .
- Find the coordinates of the point E .
- Prove that $\triangle OAD \parallel \triangle DEC$.
- Hence, or otherwise, find the ratio of the lengths AD and EC .

$$\text{(i) Gradient of } BC = \frac{6 - 0}{12 - 9} = 2.$$

$\therefore$ Gradient of $AD = 2$ (given $AD \parallel BC$).

Using $y - y_1 = m(x - x_1)$ and $A(6, 0)$,

$$\begin{aligned} y - 0 &= 2(x - 6) \\ y &= 2x - 12. \end{aligned}$$

(ii) Solve simultaneously:

$$\begin{aligned} x - 2y &= 0 \\ y &= 2x - 12. \end{aligned}$$

Substitute ② into ①:

$$\begin{aligned} x - 2(2x - 12) &= 0 \\ x - 4x + 24 &= 0 \\ x &= 8. \\ y &= 2 \times 8 - 12 \\ &= 4. \end{aligned}$$

$\therefore$ The coordinates are $D(8, 4)$.

(iii) $AB = DE = 3$ units (opposite sides of a parallelogram)

So the coordinates are $E(8 + 3, 4)$

$$\therefore E(11, 4).$$

- In $\triangle OAD$ and $\triangle DEC$,
 $\angle DOA = \angle CDE$ (corresponding $\angle$ s, $OA \parallel DE$)
 $\angle ODA = \angle DCE$ (corresponding $\angle$ s, $AD \parallel BC$)
 $\therefore \angle OAD = \angle DEC$ ($\angle$ sum of a $\triangle$)
 $\therefore \triangle OAD \parallel \triangle DEC$ (equiangular).

(v) **METHOD 1**
 $\frac{OA}{DE} = \frac{OD}{DC} = \frac{AD}{EC}$ (corresponding sides of similar $\triangle$ s are in the same ratio)

$$\begin{aligned} \frac{OA}{DE} &= \frac{6}{3} = \frac{2}{1} \\ \text{So } \frac{AD}{EC} &= \frac{2}{1} \end{aligned}$$

$$\therefore AD : EC = 2 : 1.$$

METHOD 2

Using the distance formula,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\begin{aligned} AD &= \sqrt{(8 - 6)^2 + (4 - 0)^2} \\ &= \sqrt{20} \end{aligned}$$

$$= 2\sqrt{5}.$$

$$\begin{aligned} EC &= \sqrt{(12 - 11)^2 + (6 - 4)^2} \\ &= \sqrt{5}. \end{aligned}$$

$$\therefore AD : EC = 2\sqrt{5} : \sqrt{5} = 2 : 1.$$

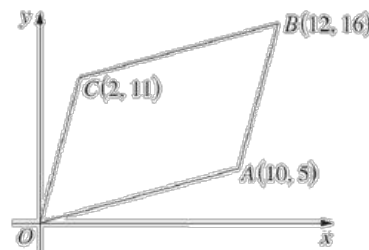


A6	2007	☒	☐
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In the diagram, A , B and C are the points $(10, 5)$, $(12, 16)$ and $(2, 11)$ respectively.

Copy or trace this diagram into your writing booklet.

- Find the distance AC .
- Find the midpoint of AC .
- Show that $OB \perp AC$.
- Find the midpoint of OB and hence explain why $OABC$ is a rhombus.
- Hence, or otherwise, find the area of $OABC$.



$$(a)(i) d_{AC} = \sqrt{(5-11)^2 + (10-2)^2} = 10$$

$$(ii) \text{midpoint} = \left(\frac{10+2}{2}, \frac{5+11}{2} \right) = (6, 8)$$

$$(iii) m_{OB} = \frac{16-0}{12-0} = \frac{4}{3}$$

$$m_{AC} = \frac{11-5}{2-10} = \frac{-3}{-8} = \frac{3}{8}$$

$$\frac{4}{3} \cdot \frac{3}{8} = 1 \therefore m_{OB} \perp m_{AC}$$

$$(iv) \text{midpoint } OB = \left(\frac{12+0}{2}, \frac{16+0}{2} \right) = (6, 8)$$

since diagonals bisect at right angles $OABC$ is rhombus

$$(v) d_{AC} = \sqrt{(16-0)^2 + (12-0)^2} = 20$$

$$A = \frac{1}{2} AC \cdot OB$$

$$A = \frac{1}{2} \cdot 10 \cdot 20 = 100 \text{ u}^2$$

A7	2008	☒	☐
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Let M be the midpoint of $(-1, 4)$ and $(5, 8)$.

Find the equation of the line through M with gradient $-\frac{1}{2}$.

$$\text{Q2bi } M\left(\frac{-1+5}{2}, \frac{4+8}{2}\right), \text{ i.e. } M(2, 6).$$

Equation of the line through M with gradient $-\frac{1}{2}$:

$$y - 6 = -\frac{1}{2}(x - 2), \quad 2y - 12 = -x + 2, \therefore x + 2y - 14 = 0.$$

A8	2011	☒	☐
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The diagram shows a line ℓ_1 , with equation $3x + 4y - 12 = 0$, which intersects the y -axis at B .

A second line ℓ_2 , with equation $4x - 3y = 0$, passes through the origin O and intersects ℓ_1 at E .

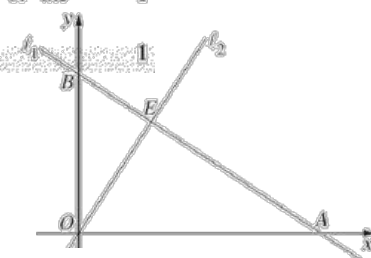
(i) Show that the coordinates of B are $(0, 3)$.

(ii) Show that ℓ_1 is perpendicular to ℓ_2 .

~~(iii) Show that the perpendicular distance from O to ℓ_1 is $\frac{12}{5}$.~~

(iv) Using Pythagoras' theorem, or otherwise, find the length of the interval BE .

~~(v) Hence, or otherwise, find the area of $\triangle BOE$.~~



$$\text{Q3ci } \ell_1: 3x + 4y - 12 = 0$$

$$\text{Let } x = 0, y = 3, \therefore B(0, 3)$$

$$\text{Q3cii } m_1 = -\frac{3}{4}, m_2 = \frac{4}{3}, m_1 m_2 = -1, \therefore \ell_1 \text{ and } \ell_2 \text{ are } \perp$$

$$\text{Q3ciii } \ell_1: 3x + 4y - 12 = 0. \text{ Let } y = 0, x = 4, \therefore A(4, 0)$$

$$\therefore AB = \sqrt{3^2 + 4^2} = 5$$

$$\triangle OAB \text{ and } \triangle OEB, \therefore \frac{OE}{OA} = \frac{OB}{AB}, \frac{OE}{4} = \frac{3}{5}, OE = \frac{12}{5}$$

$$\text{Q3civ } \frac{BE}{BO} = \frac{OE}{OA}, \frac{BE}{3} = \frac{\frac{12}{5}}{4}, BE = \frac{9}{5}$$

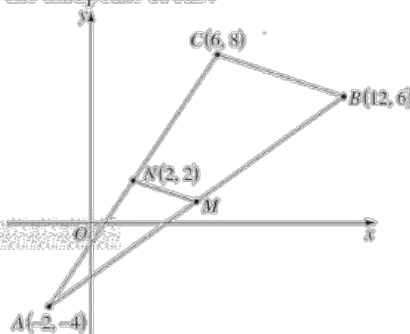
$$\text{Q3cv } \text{Area of } \triangle BOE = \frac{1}{2} \times \frac{12}{5} \times \frac{9}{5} = \frac{54}{25} \text{ unit squares}$$



A9	2010	☒	☐
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In the diagram A , B and C are the points $(-2, -4)$, $(12, 6)$ and $(6, 8)$ respectively. The point $N(2, 2)$ is the midpoint of AC . The point M is the midpoint of AB .

- Find the coordinates of M .
- Find the gradient of BC .
- Prove that $\triangle ABC$ is similar to $\triangle AMN$.
- Find the equation of MN .
- Find the exact length of BC .
- Given that the area of $\triangle ABC$ is 44 square units, find the perpendicular distance from A to BC .



Q3ai Midpoint $M\left(\frac{-2+12}{2}, \frac{-4+6}{2}\right) = M(5, 1)$

Q3aii Gradient of $BC = \frac{8-6}{6-12} = -\frac{1}{3}$

Q3aiii Gradient of $MN = \frac{2-1}{2-5} = -\frac{1}{3}$

$\therefore MN$ and BC are parallel

$\therefore$ corresponding angles $\angle AMN = \angle ABC$ and corresponding angles $\angle ANM = \angle ACB$

Hence $\triangle AMN, \triangle ABC$ are similar.

Q3aiv Equation of MN : $y - 2 = -\frac{1}{3}(x - 2)$, $y = -\frac{1}{3}x + \frac{8}{3}$ or $x + 3y = 8$

Q3av Length of $BC = \sqrt{(12-6)^2 + (6-8)^2} = \sqrt{40} = 2\sqrt{10}$

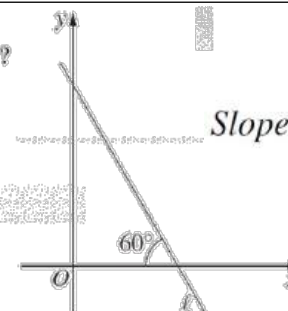
Q3avi Let h be the perpendicular distance from A to BC .

$\frac{1}{2} \times 2\sqrt{10} \times h = 44$, $h = \frac{44}{\sqrt{10}} = \frac{22\sqrt{10}}{5}$

A10	2013	☒	☐
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The diagram shows the line ℓ . What is the slope of the line ℓ ?

- $\sqrt{3}$
- $-\sqrt{3}$
- $\frac{1}{\sqrt{3}}$
- $-\frac{1}{\sqrt{3}}$



$Slope = \tan(180^\circ - 60^\circ) = -\sqrt{3}$

A11	2015	☒	☐
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What is the slope of the line with equation $2x - 4y + 3 = 0$?

- -2
- $-\frac{1}{2}$
- $\frac{1}{2}$
- 2

$2x - 4y + 3 = 0 \Rightarrow y = \frac{1}{4}(2x + 3) \therefore \text{slope} = \frac{1}{2}$

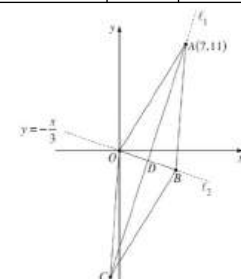
A12	2015	☒	☐
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The diagram shows the rhombus $OABC$.

The diagonal from the point $A(7, 11)$ to the point C lies on the line ℓ_1 .

The other diagonal, from the origin O to the point B , lies on the line ℓ_2 which has equation $y = -\frac{x}{3}$.

- Show that the equation of the line ℓ_1 is $y = 3x - 10$. 2
- The lines ℓ_1 and ℓ_2 intersect at the point D . Find the coordinates of D . 2



(ii) Diagonals of rhombus intersect at right angles, thus $m_{\ell_1} = 3$. (iii) Solving the set of simultaneous equations:

By point gradient formula,

$\frac{y-11}{x-7} = 3$

$y - 11 = 3x - 21$

$\therefore y = 3x - 10$

$\begin{cases} y = 3x - 10 \\ y = -\frac{x}{3} \end{cases}$

$\Rightarrow 3x - 10 = -\frac{x}{3}$

$10x - 30 = 0$

$\therefore x = 3, y = -1$

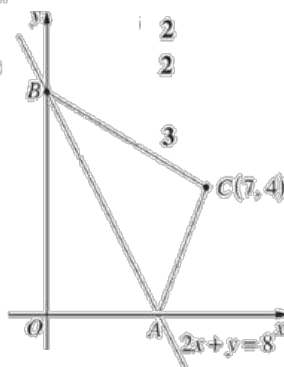
$\therefore D(3, -1)$



A13	2012	☒	☐
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The diagram shows a triangle ABC . The line $2x + y = 8$ meets the x and y axes at the points A and B respectively. The point C has coordinates $(7, 4)$.

- Calculate the distance AB .
- It is known that $AC = 5$ and $BC = \sqrt{65}$. (Do NOT prove this.) Calculate the size of $\angle ABC$ to the nearest degree.
- The point N lies on AB such that CN is perpendicular to AB . Find the coordinates of N .



Q13ai $A(4, 0)$, $B(0, 8)$, $\therefore AB = \sqrt{(0-4)^2 + (8-0)^2} = 4\sqrt{5}$

Q13aii The cosine rule: $b^2 = c^2 + a^2 - 2ac \cos \angle ABC$

$$5^2 = (4\sqrt{5})^2 + (\sqrt{65})^2 - 2(4\sqrt{5})(\sqrt{65}) \cos \angle ABC$$

$$25 = 80 + 65 - 8\sqrt{325} \cos \angle ABC, \cos \angle ABC = \frac{80 + 65 - 25}{8\sqrt{325}},$$

$$\therefore \angle ABC \approx 34^\circ$$

Q13aiii Let $N(p, q)$. $\therefore 2p + q = 8$ (1)

$$m_{CN} \times m_{AB} = -1, \therefore \frac{q-4}{p-7} \times -2 = -1 \text{ (2)}$$

From (1), $q = 8 - 2p$ (3)

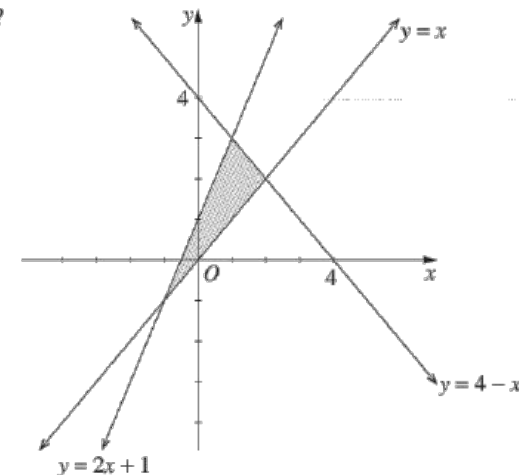
Substitute (3) in (2), $\frac{8-2p-4}{p-7} \times 2 = 1, \therefore 8-4p = p-7,$

$$\therefore p = 3 \text{ and } q = 2, \therefore N(3, 2)$$

A14	2017	☒	☐
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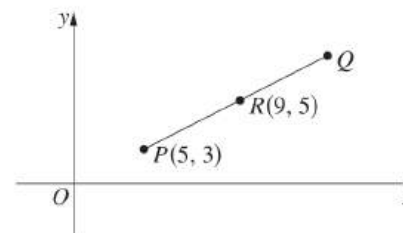
The region enclosed by $y = 4 - x$, $y = x$ and $y = 2x + 1$ is shaded in the diagram below. Which of the following defines the shaded region?

- $y \leq 2x + 1, y \leq 4 - x, y \geq x$
- $y \geq 2x + 1, y \leq 4 - x, y \geq x$
- $y \leq 2x + 1, y \geq 4 - x, y \geq x$
- $y \geq 2x + 1, y \geq 4 - x, y \geq x$



A15	2018	☒	☐
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The point $R(9, 5)$ is the midpoint of the interval PQ , where P has coordinates $(5, 3)$.



NOT TO
SCALE

What are the coordinates of Q ?

- $(4, 7)$
- $(7, 4)$
- $(13, 7)$
- $(14, 8)$



A16	2019	☒	☐
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The line ℓ , with equation $x - 2y + 4 = 0$, passes through the point $A(8, 6)$ and intersects the x -axis at the point B .

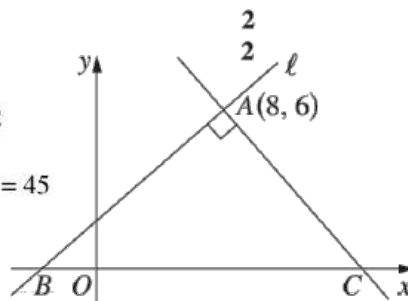
The line through A perpendicular to ℓ intersects the x -axis at the point C .

(i) Find the equation of the line AC .

(ii) Find the area of triangle ABC .

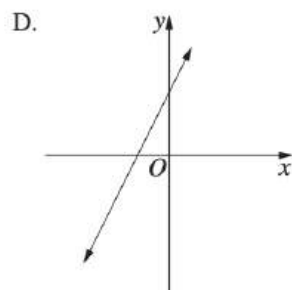
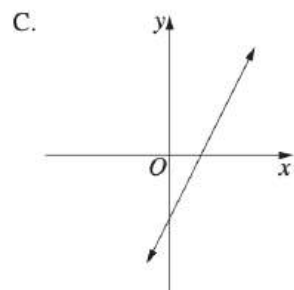
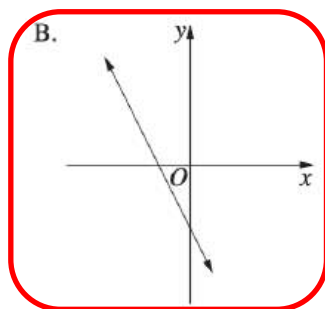
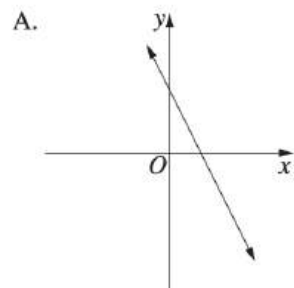
i Gradient of $\ell = \frac{1}{2}$, line AC $\frac{y-6}{x-8} = -2$, $y = -2x + 22$

ii $B(-4, 0)$, $C(11, 0)$, area $\triangle ABC = \frac{1}{2}(11 - (-4)) \times 6 = 45$



A17	2022	☒	☐
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Which of the following could be the graph of $y = -2x + 2$?



A18	2020	☒	☐
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There are two tanks on a property, Tank A and Tank B. Initially, Tank A holds 1000 litres of water and Tank B is empty.

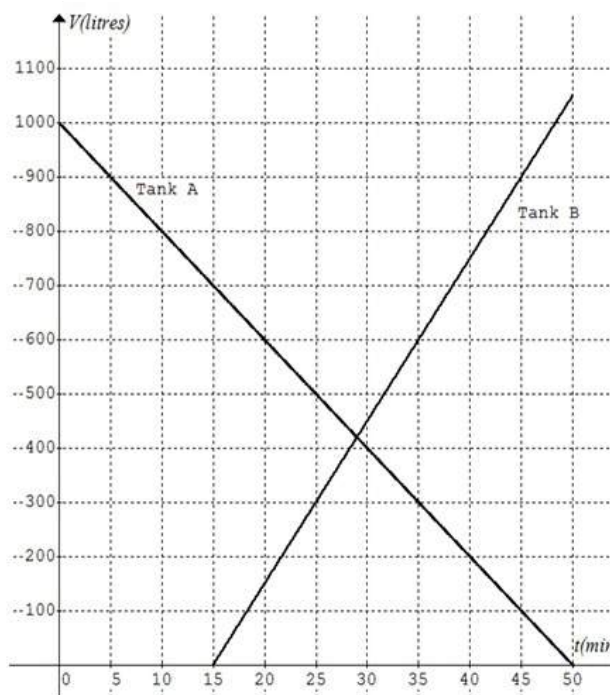
(a) Tank A begins to lose water at a constant rate of 20 litres per minute. The volume of water in Tank A is modelled by $V = 1000 - 20t$ where V is the volume in litres and t is the time in minutes from when the tank begins to lose water. 1

On the grid below, draw the graph of this model and label it as Tank A.

(b) Tank B remains empty until $t = 15$ when water is added to it at a constant rate of 30 litres per minute. 2

By drawing a line on the grid on the previous page, or otherwise, find the value of t when the two tanks contain the same volume of water.

(c) Using the graphs drawn, or otherwise, find the value of t (where $t > 0$) when the total volume of water in the two tanks is 1000 litres. 1



b Tank A: $V = 1000 - 20t$;
Tank B: $V = 30(t - 15)$
Same volume: $30(t - 15) = 1000 - 20t$, $50t = 1450$,
 $t = 29$

c $30(t - 15) + 1000 - 20t = 1000$, $t > 0 \therefore 10t = 450$,
 $t = 45$



A19	2024	☒	☐
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Consider the function shown.

Which of the following could be the equation of this function?

A. $y = 2x + 3$

B. $y = 2x - 3$

C. $y = -2x + 3$

D. $y = -2x - 3$

